

Multivariable Calculus

Unit 2
Jaysen Tsao

2	Vector Valued Functions	2
2.1	Vector Functions	2
	Space Curves	2
	Intersections of Planes	2
2.2	Calculus with Vector Functions	3
	Derivatives of Vector Functions	3
	Differentiation Rules for Vector Functions	4
	Integrals of Vector Functions	4
2.3	Arc Length and Curvature	5
	Arc Length	5
	Parameterizations	5
	Curvature	6
	Normal Vectors and Planes	9
	Extension: Torsion	11
	Summary	12
2.4	Applications of Vector Functions	13
	Particle Motion Definitions	13
	Tangential and Centripetal Acceleration	13
2.5	Parametric Surfaces	15
	Grid Curves	15
	Parametric Equations of a Plane	15

2 Vector Valued Functions

2.1 Vector Functions

- We've looked at functions which took n inputs and output one value, called scalar functions of n variables. We also said that these functions satisfy the mapping $\mathbb{R}^n \mapsto \mathbb{R}$.
- **Vector functions** (or vector-valued functions) are functions which take a certain amount of variables and output *many* values. These values are represented as vectors.

We say that a vector function $\vec{r}$ of m variables which outputs n values satisfies the mapping $\mathbb{R}^m \mapsto \mathbb{R}^n$. We will only be dealing with vector functions of one variable: $\vec{r}: \mathbb{R} \mapsto \mathbb{R}^n$.

- The components of a vector function $\vec{r}$ are known as the **component functions** of $\vec{r}$. For example, f , g , and h are the component functions of $\vec{r}$ if $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$.

Domain of a Vector Function

The domain D of a vector function $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$ is the **intersection** of the domains of $f_1, f_2, \dots, f_n$ in t :

$$D = \text{Dom}(\vec{r}(t)) = \bigcap_{k=1}^n \text{Dom}(f_k(t)).$$

Limit of a Vector Function

The limit of a vector function of one variable $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$ is the vector containing the limits of each component function of $\vec{r}$:

$$\lim_{t \rightarrow c} \vec{r}(t) = \left\langle \lim_{t \rightarrow c} f_1(t), \lim_{t \rightarrow c} f_2(t), \dots, \lim_{t \rightarrow c} f_n(t) \right\rangle.$$

Space Curves

Vector functions of one variable trace out **space curves** in 3-space. If all component functions of a vector function $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$ are continuous on some interval I , then we can define a **space curve** C as the set of points represented by the position vectors $\forall t \in I$.

The space curve C will have parametric equations: $x = f(t)$, $y = g(t)$, $z = h(t)$, and this can be shown by simply assuming $\vec{r}(t)$ represents the position vector of a point on C at any time t :

$$\vec{r}(t) = \langle x, y, z \rangle = \langle f(t), g(t), h(t) \rangle.$$

Intersections of Planes

- **The intersection of two surfaces is a space curve.** We know that two planes in 3-space can have a *line of intersection*. In a similar vein, any two surfaces in 3-space can have a **curve of intersection**, which is a space curve.

2.2 Calculus with Vector Functions

Derivatives of Vector Functions

The derivative $\vec{r}'(t)$ of a vector function $\vec{r}(t)$ is defined the same way as for normal functions:

$$\frac{d\vec{r}}{dt} = \vec{r}'(t) = \lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h}.$$

Derivation: Derivative of a Vector Function by Components

Using basic laws of vectors and limits, we can also see that the derivative of a vector function is just a vector of the derivatives of each component function. If $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$, then:

$$\begin{aligned} \vec{r}'(t) &= \lim_{h \rightarrow 0} \frac{\langle f_1(t+h), f_2(t+h), \dots, f_n(t+h) \rangle - \langle f_1(t), f_2(t), \dots, f_n(t) \rangle}{h} \\ &= \lim_{h \rightarrow 0} \frac{\langle f_1(t+h) - f_1(t), f_2(t+h) - f_2(t), \dots, f_n(t+h) - f_n(t) \rangle}{h} \\ &= \lim_{h \rightarrow 0} \left\langle \frac{f_1(t+h) - f_1(t)}{h}, \frac{f_2(t+h) - f_2(t)}{h}, \dots, \frac{f_n(t+h) - f_n(t)}{h} \right\rangle \\ &= \left\langle \lim_{h \rightarrow 0} \frac{f_1(t+h) - f_1(t)}{h}, \lim_{h \rightarrow 0} \frac{f_2(t+h) - f_2(t)}{h}, \dots, \lim_{h \rightarrow 0} \frac{f_n(t+h) - f_n(t)}{h} \right\rangle \\ &= \langle f_1'(t), f_2'(t), \dots, f_n'(t) \rangle. \end{aligned}$$

Derivative of a Vector Function

Let the vector function $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$ have differentiable components. Then the derivative of $\vec{r}$, $\vec{r}'$ is a vector of the derivatives of each component:

$$\frac{d}{dt} \vec{r}(t) = \vec{r}'(t) = \langle f_1'(t), f_2'(t), \dots, f_n'(t) \rangle.$$

Similarly, the k th derivative of $\vec{r}$ is:

$$\frac{d^k}{dt^k} \vec{r}(t) = \vec{r}^{(k)}(t) = \langle f_1^{(k)}(t), f_2^{(k)}(t), \dots, f_n^{(k)}(t) \rangle$$

Remember that in 3-space, if the component functions of $\vec{r}(t)$ are continuous over an interval I , there exists a **space curve** C represented by $\vec{r}(t)$. For any given value of $t \in I$, $\vec{r}'(t)$ represents a vector tangent to C at the position represented by $\vec{r}(t)$. This vector is called the **tangent vector**.

Similarly, the **unit tangent vector** is often used to evaluate which direction the curve is moving in at any given $t \in I$, but not necessarily at what magnitude. This vector is represented:

$$\text{Unit Tangent Vector} = \vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}.$$

Differentiation Rules for Vector Functions

If $\vec{u}(t)$ and $\vec{v}(t)$ are differentiable vector functions, then:

- Linearity Rule: $\frac{d}{dt}(a\vec{u}(t) + b\vec{v}(t)) = a\vec{u}'(t) + b\vec{v}'(t)$ where a and b are constants.
- Scalar Product Rule: $\frac{d}{dt}f(t)\vec{u}(t) = f'(t)\vec{u}(t) + f(t)\vec{u}'(t)$ where $f(t)$ is a scalar function.
- Dot Product Rule: $\frac{d}{dt}(\vec{u}(t) \cdot \vec{v}(t)) = \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t)$.
- Cross Product Rule: $\frac{d}{dt}(\vec{u}(t) \times \vec{v}(t)) = \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t)$.
- Chain Rule: $\frac{d}{dt}\vec{u}(f(t)) = \vec{u}'(f(t))f'(t)$ where $f(t)$ is a scalar function.

Integrals of Vector Functions

Definite Integral of a Vector Function

The definite integral from $t = a$ to $t = b$ of a vector function $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$ is the vector consisting of the definite integrals from $t = a$ to $t = b$ over each of its components:

$$\int_a^b \vec{r}(t) dt = \left\langle \int_a^b f_1(t) dt, \int_a^b f_2(t) dt, \dots, \int_a^b f_n(t) dt \right\rangle.$$

Antiderivative of a Vector Function

The antiderivative $\vec{R}(t)$ of a vector function $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$ is one where $\vec{R}'(t) = \vec{r}(t)$. It is the vector consisting of the antiderivatives of component functions, but it can vary by a constant vector $\vec{C}$ (since each component can vary independently):

$$\vec{R}(t) = \int \vec{r}(t) dt = \left\langle \int f_1(t) dt, \int f_2(t) dt, \dots, \int f_n(t) dt \right\rangle + \vec{C}.$$

2.3 Arc Length and Curvature

Arc Length

Derivation: Arc Length of a Space Curve

Let's derive the arc length s of a space curve C in $\mathbb{R}^3$ defined by $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$ from $t = a$ to $t = b$.

We know that an infinitesimally small movement along the curve C , $d\vec{r}$, is represented by their respective movements in the x , y , and z -axes:

$$d\vec{r} = \langle dx, dy, dz \rangle.$$

The magnitude of each small movement $d\vec{r}$ results in a small contribution to arc length:

$$\begin{aligned} ds = |d\vec{r}| &= \sqrt{dx^2 + dy^2 + dz^2} \\ &= \sqrt{\left(\frac{dx}{dt} dt\right)^2 + \left(\frac{dy}{dt} dt\right)^2 + \left(\frac{dz}{dt} dt\right)^2} \\ &= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \end{aligned}$$

We can finally add up all small changes in arc length from $t = a$ to $t = b$ using an integral:

$$s = \int_a^b ds = \int_a^b |\vec{r}'(t)| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

Arc Length of a Space Curve

Let the space curve C be defined by the vector function $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$. Then the **arc length** s of C from $t = a$ to $t = b$ is:

$$\text{Arc Length} = s = \int_a^b |\vec{r}'(t)| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt.$$

Parameterizations

- We know that a space curve C can be defined by a vector-valued function represented by a set of parametric equations, but there exists an infinite set of parametric equations for C called the **parameterizations** of C .

For example, let the curve C be defined using the vector function $\vec{r}(t) = \langle t, t^2, \sin t \rangle$. We can apply the transformation $t = e^u$ to get a new curve D , $\vec{r}(u) = \langle e^u, e^{2u}, \sin(e^u) \rangle$. Since there exists a bijection from t to e^u , all points on C are also points on D , and vice-versa, so $C \equiv D$.

- We can often make the definition of a curve independent of parameterization by defining it in terms of “how far we are along the curve”, i.e. arc length:

Arc Length Function of a Curve

If the curve C is defined using the vector function $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$, its **arc length function** $\vec{R}(s)$ is parameterization of $\vec{r}$ using s , where $s(t)$ is the arc length from some starting point $t = a$ to t :

$$s(t) = \int_a^t |\vec{r}'(\tau)| d\tau = \int_a^t \sqrt{\left(\frac{dx}{d\tau}\right)^2 + \left(\frac{dy}{d\tau}\right)^2 + \left(\frac{dz}{d\tau}\right)^2} d\tau.$$

It holds that $\frac{ds}{dt} = |\vec{r}'(t)|$ by the fundamental theorem of calculus. We can use this differential equation to determine a transformation from $t \rightarrow s$.

Example: Reparameterization using Arc Length

The curve C is defined by $\vec{r}(t) = \langle 3t, 5 - 4t, 6 + 2t \rangle$.

Write an *arc length function* $\vec{q}(s)$ for C .

First, we know that $\vec{r}'(t) = \langle 3, -4, 2 \rangle$ and thus $|\vec{r}'(t)| = \sqrt{29}$.

We can find the arc length s as a function of t , choosing $s = 0$ at $t = 0$:

$$s(t) = \int_0^t |\vec{r}'(\tau)| d\tau = \int_0^t \sqrt{29} d\tau = \sqrt{29}t$$
$$\Rightarrow t = \frac{s}{\sqrt{29}}.$$

After solving for t in terms of s , we perform the substitution, and the parameterization is:

$$\vec{q}(s) = \left\langle \frac{3}{\sqrt{29}}s, 5 - \frac{4}{\sqrt{29}}s, 6 + \frac{2}{\sqrt{29}}s \right\rangle.$$

Curvature

Curvature is a measure of how quickly a space curve changes direction at any given point. The measure is not concerned about the magnitude of change; only direction.

Recall that the direction of the tangent vector can be represented using the **unit tangent vector**, $\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$. Then the curvature κ is how quickly this vector changes with respect to where we are on the curve, $\kappa = \left| \frac{d\vec{T}}{ds} \right|$.

Different parameterizations of t can produce the same curve (for example $\langle t, 2t, 3t \rangle$ and $\langle 2t, 4t, 6t \rangle$), but the progression along the curve may be slower or faster if t increases at given rate. Thus, we use the *arc length* as a measure of how far we are on the curve.

Curvature of a Space Curve

Let the space curve C be defined by the vector function $\vec{r}(t)$. If an arc length parameterization $\vec{T}(s)$ of the unit tangent vector exists, then the **curvature** κ of C is:

$$\text{Curvature} = \kappa = \left| \frac{d\vec{T}}{ds} \right|.$$

If $\vec{T}(t)$ is the unit vector tangent to $\vec{r}(t)$ at t , then the curvature κ of C at t is:

$$\kappa(t) = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3}.$$

Special Case: Curvature of 2D Curve

If C completely lies on a plane and can be defined by $y = f(x)$, then we can parameterize this into a vector valued function $\vec{r}(x) = \langle x, f(x), 0 \rangle$.

Then $\vec{r}'(x) = \langle 1, f'(x), 0 \rangle$, $\vec{r}''(x) = \langle 0, f''(x), 0 \rangle$, and $\vec{r}'(x) \times \vec{r}''(x) = \langle 0, 0, f''(x) \rangle$:

$$\kappa(x) = \frac{|f''(x)|}{(1 + (f'(x))^2)^{3/2}}.$$

Example: Computing the Curvature of a Space Curve

Find the curvature of the space curve defined by $\vec{r}(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ as a function of t .

Since we are only given $\vec{r}$, it would be most convenient to use the formula in terms of only $\vec{r}$ and its derivatives.

Compute the derivatives of $\vec{r}$:

$$\vec{r}(t) = \langle t, t^2, t^3 \rangle$$

$$\vec{r}'(t) = \langle 1, 2t, 3t^2 \rangle$$

$$\vec{r}''(t) = \langle 0, 2, 6t \rangle.$$

Compute $|\vec{r}' \times \vec{r}''|$:

$$\begin{aligned} |\vec{r}' \times \vec{r}''| &= \left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2t & 3t^2 \\ 0 & 2 & 6t \end{vmatrix} \right\| \\ &= |\langle 6t^2, -6t, 2 \rangle| \\ &= \sqrt{36t^4 + 36t^2 + 4} \\ &= 2\sqrt{9t^4 + 9t^2 + 1}. \end{aligned}$$

Compute $|\vec{r}'|^3$:

$$\begin{aligned} |\vec{r}'|^3 &= |\langle 1, 2t, 3t^2 \rangle|^3 \\ &= \left| \sqrt{1 + 4t^2 + 9t^4} \right|^3 \\ &= (1 + 4t^2 + 9t^4)^{3/2}. \end{aligned}$$

The quotient is the curvature:

$$\kappa = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3} = \frac{2\sqrt{9t^4 + 9t^2 + 1}}{(1 + 4t^2 + 9t^4)^{3/2}}.$$

Derivation: Curvature as functions of t

By the chain rule:

$$\kappa = \left| \frac{d\vec{T}}{ds} \right| = \left| \frac{d\vec{T}/dt}{ds/dt} \right| = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|}.$$

By the quotient rule:

$$\vec{T}'(t) = \frac{\vec{r}''(t)|\vec{r}'(t)| - \vec{r}'(t)\frac{d}{dt}|\vec{r}'(t)|}{|\vec{r}'(t)|^2}.$$

Recall that $\frac{d|x|}{dx} = \frac{x}{|x|}$. The same can be applied to vector magnitudes:

$$\begin{aligned} \vec{T}'(t) &= \frac{\vec{r}''(t)|\vec{r}'(t)| - \vec{r}'(t)\frac{r'(t)}{|\vec{r}'(t)|} \cdot \vec{r}'(t)}{|\vec{r}'(t)|^2} \\ &= \frac{\vec{r}''(t)|\vec{r}'(t)|^2 - \vec{r}'(t)(\vec{r}'(t) \cdot \vec{r}''(t))}{|\vec{r}'(t)|^3} \end{aligned}$$

Recall the a property of vector triple products, $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$.

Since $|\vec{r}'(t)|^2 = \vec{r}'(t) \cdot \vec{r}'(t)$:

$$\begin{aligned} \vec{r}''(t)|\vec{r}'(t)|^2 - \vec{r}'(t)(\vec{r}'(t) \cdot \vec{r}''(t)) &= (\vec{r}'(t) \cdot \vec{r}'(t))\vec{r}''(t) - (\vec{r}'(t) \cdot \vec{r}''(t))\vec{r}'(t) \\ &= \vec{r}'(t) \times (\vec{r}''(t) \times \vec{r}'(t)). \end{aligned}$$

Since $\vec{r}''(t) \times \vec{r}'(t)$ is orthogonal to $\vec{r}'(t)$, the following holds:

$$\begin{aligned} |\vec{r}'(t) \times (\vec{r}''(t) \times \vec{r}'(t))| &= |\vec{r}'(t)| |\vec{r}''(t) \times \vec{r}'(t)| \\ &= |\vec{r}'(t)| |\vec{r}'(t) \times \vec{r}''(t)|. \end{aligned}$$

Finally,

$$\begin{aligned} |\vec{T}'(t)| &= \frac{|\vec{r}'(t)| |\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^2}. \\ \kappa(t) &= \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^2} \frac{1}{|\vec{r}'(t)|} = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3}. \end{aligned}$$

Normal Vectors and Planes

- We know that $\vec{T}$, the unit tangent vector, will always have a magnitude of 1. Thus, $|\vec{T}|$ is always equal to 1. Since $|\vec{T}|$ is constant, then $|\vec{T}'| = 0$ and $\vec{T} \cdot \vec{T}' = |\vec{T}| |\vec{T}'| \cos \theta = 0$. Thus, $\vec{T}$ is always orthogonal to its derivative $\vec{T}'$.

The unit vector of $\vec{T}'$ is known as the **principal unit normal vector**, $\vec{N}(t) = \frac{\vec{T}'(t)}{|\vec{T}'(t)|}$.

Proof: A vector function $\vec{r}$ is orthogonal to $\vec{r}'$ if $\vec{r}$ has a constant length

If the statement above can be proved true, then we can be certain that because the unit tangent vector $\vec{T}$ has a constant length 1, it must be orthogonal to $\vec{T}'$, and since $\vec{N}$ is parallel to $\vec{T}'$, $\vec{N}$ is orthogonal to $\vec{T}$ and thus $\vec{r}$.

Let $\vec{r}(t)$ be a vector function in n -space with components $\vec{r}(t) = \langle f_1(t), f_2(t), \dots, f_n(t) \rangle$. Then the tangent vector function is $\vec{r}'(t) = \langle f_1'(t), f_2'(t), \dots, f_n'(t) \rangle$.

If $\vec{r}(t)$ has a constant magnitude, then $f_1^2(t) + f_2^2(t) + \dots + f_n^2(t) = c$, where c is a constant.

By implicit differentiation:

$$\begin{aligned} f_1^2(t) + f_2^2(t) + \dots + f_n^2(t) &= c \\ 2f_1(t)f_1'(t) + 2f_2(t)f_2'(t) + \dots + 2f_n(t)f_n'(t) &= 0 \\ f_1(t)f_1'(t) + f_2(t)f_2'(t) + \dots + f_n(t)f_n'(t) &= 0 \\ \vec{r}(t) \cdot \vec{r}'(t) &= 0. \quad \blacksquare \end{aligned}$$

This is exactly the dot product of $\vec{r}(t)$ and $\vec{r}'(t)$. The dot product of two orthogonal vectors is 0, so $\vec{r}$ must be orthogonal to $\vec{r}'$ at all times if $\vec{r}$ has a constant magnitude.

- The **binormal vector** $\vec{B}$ is the unit vector orthogonal to both $\vec{T}$ and $\vec{N}$ at the same time, so it can be represented by $\vec{B}(t) = \vec{T}(t) \times \vec{N}(t)$. Since both $\vec{T}$ and $\vec{N}$ are unit vectors, $\vec{B}$ is also a unit vector.

The TNB Frame

A **normal unit vector** $\vec{n}$ to a space curve $\vec{r}(t)$ at t is orthogonal to $\vec{T}(t) = \frac{\vec{r}'(t)}{|\vec{r}'(t)|}$, the unit tangent vector at t . We typically deal with two normal unit vectors: the **principal normal unit vector** or **unit normal** $\vec{N}(t)$ and the **binormal vector** $\vec{B}(t)$, where:

$$\vec{N}(t) = \frac{\vec{T}'(t)}{|\vec{T}'(t)|} \quad \vec{B}(t) = \vec{T}(t) \times \vec{N}(t).$$

The **TNB frame** is the set of unit vectors $\vec{T}$, $\vec{N}$, and $\vec{B}$ which define how a curve is behaving at a point, which are all components of the derivatives of $\vec{r}$, where:

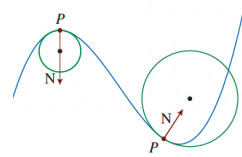
- $\vec{T}$ represents how the particle is *moving*, quantified using *speed* $v = \left| \frac{d\vec{r}}{dt} \right|$
- $\vec{N}$ represents how the particle is *turning*, quantified using *curvature*, $\kappa = \left| \frac{d\vec{T}}{ds} \right|$
- $\vec{B}$ represents how the particle is *twisting*, quantified using *torsion*, $\tau = -\frac{d\vec{B}}{ds} \cdot \vec{N}$.

- The plane P which contains both $\vec{N}$ and $\vec{B}$ is called the **normal plane** because it is normal to the tangent vector $\vec{T}$, meaning the vector normal to P is parallel to $\vec{T}$.
- The **rectifying plane** R is the plane which contains both $\vec{T}$ and $\vec{B}$, lying normal to $\vec{N}$.
- The **osculating plane** O is the plane that contains both $\vec{T}$ and $\vec{N}$, and is the plane which lies close to the curve near the point of tangency. The osculating plane is normal to $\vec{B}$.
- If we know the tangent and normal vectors to a space curve C at a point P , we can determine the **circle of curvature** or **osculating circle** at P , which approximates the behavior C using a

circle (as opposed to a tangent line). They can be useful as circles of curvature consider the curvature κ in the approximation:

Circle of Curvature

Let the space curve C have curvature κ at a point P . Then the **circle of curvature** or **osculating circle** at P is the circle which (1) lies on the osculating plane of C , (2) contains the point P , (3) has radius $R = \frac{1}{\kappa}$, and (4) has a center Q a distance $\frac{1}{\kappa}$ from P along the principal normal vector $\vec{N}$:



$$\vec{Q} = \vec{P} + \frac{\vec{N}}{\kappa} \quad R = \frac{1}{\kappa}$$

R is known as the **radius of curvature**.

Extension: Torsion

Torsion is a quantity τ which measures how fast the osculating plane rotates as we move along the curve (i.e. with respect to arc length s). Since the osculating plane is described using its normal vector $\vec{B}$, we can suppose that τ should be expressed in terms of $\frac{d\vec{B}}{ds}$.

Torsion of a Space Curve

Let a space curve $\vec{r}$ have the TNB unit vectors $\vec{T}, \vec{N}, \vec{B}$ at some point P on the curve. Then the **torsion** τ at P is:

$$\text{Torsion} = \tau = -\frac{d\vec{B}}{ds} \cdot \vec{N}.$$

With Respect to Time

The torsion can also be found using the following formula:

$$\tau(t) = \frac{\det(\vec{r}, \vec{r}', \vec{r}'')}{|\vec{r}' \times \vec{r}''|^2}.$$

Example: Torsion of a Circular Helix

Find the torsion of the circular helix $\vec{r}(t) = \langle a \cos t, a \sin t, bt \rangle$ as a function of t , where a and b are constants.

First, find relevant derivatives:

$$\vec{r}' = \langle -a \sin t, a \cos t, b \rangle$$

$$\vec{r}'' = \langle -a \cos t, -a \sin t, 0 \rangle$$

$$\vec{r}''' = \langle a \sin t, -a \cos t, 0 \rangle.$$

Then, $\det(\vec{r}', \vec{r}'', \vec{r}''')$ is the same as:

$$\begin{vmatrix} -a \sin t & a \cos t & b \\ -a \cos t & -a \sin t & 0 \\ a \sin t & -a \cos t & 0 \end{vmatrix} = a^2 b.$$

Find: $|\vec{r}' \times \vec{r}''|^2$:

$$\begin{aligned} |\vec{r}' \times \vec{r}''|^2 &= \left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -a \sin t & a \cos t & b \\ -a \cos t & -a \sin t & 0 \end{vmatrix} \right\|^2 = |\langle ab \sin t, ab \cos t, a^2 \sin^2 t + a^2 \cos^2 t \rangle|^2 \\ &= |\langle ab \sin t, ab \cos t, a^2 \rangle|^2 \\ &= a^2 b^2 \sin^2 t + a^2 b^2 \cos^2 t + a^4 = a^2(b^2 + 1). \end{aligned}$$

Take the quotient to find torsion:

$$\tau = \frac{a^2 b}{a^2(b^2 + 1)} = \frac{b}{b^2 + 1}.$$

Summary

Arc Length: $s = \int_a^b |\vec{r}'(t)| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$

Curvature (3D): $\kappa = \left| \frac{d\vec{T}}{ds} \right| = \frac{|\vec{T}'(t)|}{|\vec{r}'(t)|} = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}$

Curvature (2D): $\kappa = \frac{|f''(x)|}{(1 + (f'(x))^2)^{3/2}}$

Radius of Curvature: $r = \frac{1}{\kappa}$

TNB Frame: $\vec{T} = \frac{\vec{r}'}{|\vec{r}'|} \quad \vec{N} = \frac{\vec{T}'}{|\vec{T}'|} \quad \vec{B} = \vec{T} \times \vec{N}$

$\vec{T}$ normal to normal plane; $\vec{N}$ normal to rectifying plane; $\vec{B}$ normal to osculating plane

Torsion: $\tau = -\frac{d\vec{B}}{ds} \cdot \vec{N} = \frac{\det(\vec{r}', \vec{r}'', \vec{r}''')}{|\vec{r}' \times \vec{r}''|}$

2.4 Applications of Vector Functions

Particle Motion Definitions

If $\vec{r}(t)$ is the **position vector** of a particle at any time t , then:

- $\vec{v}(t) = \vec{r}'(t)$ is the **velocity vector** at time t
 - Its magnitude $v(t) = |\vec{v}(t)| = |\vec{r}'(t)|$ is the **speed** at time t
 - We can also take the derivative of *distance traveled*, s , over time: $v(t) = s'(t) = \frac{d}{dt}|\vec{r}(t)|$.
From this, we can see that $|\frac{d}{dt}\vec{r}(t)| = \frac{d}{dt}|\vec{r}(t)|$. This is a special property of the position vector.
- $\vec{a}(t) = \vec{v}'(t) = \vec{r}''(t)$ is the **acceleration vector** at time t .

Tangential and Centripetal Acceleration

- We have been able to come up with a scalar analog of velocity, *speed*, which is the derivative of arc length or the magnitude of velocity.
- However, the same cannot be said for acceleration. The derivative of speed is not necessarily equal to the magnitude of acceleration. In other words, $|\vec{v}'(t)| \neq v'(t)$.
- When we only consider speed, the changing direction of the velocity vector is lost. The function $\vec{v}(t)$ would be the same whether we were moving straight at a constant rate or whether we are moving around in a circle at a constant rate.
- Thus, we must consider acceleration using at least two scalar components in order to properly represent the potential to change direction. When evaluating particle dynamics, it is useful to break down acceleration into two useful components:
 - **tangential acceleration**, a_t , which is the component of acceleration directed tangent to the curve (in the same direction as $\vec{T}$)
 - **centripetal acceleration**, a_c , which is the component of acceleration responsible for changing the direction of the particle, directed towards the center of the circle of curvature, which will always be normal to the curve (in the same direction as $\vec{N}$).

Components of Acceleration

The acceleration vector $\vec{a}$ can be broken down into two components.

- **Tangential Acceleration** a_t is the derivative of speed, $a_t = \frac{dv}{dt} = \frac{d^2s}{dt^2} = \frac{\vec{r}' \cdot \vec{r}''}{|\vec{r}'|}$.
- **Centripetal Acceleration** a_c is $a_c = \kappa v^2 = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}$, where κ is curvature.

If $\vec{T}$ is the unit tangent vector to the curve at t and $\vec{N}$ is the unit normal at t , then:

$$\vec{a} = a_t \vec{T} + a_c \vec{N} = v' \vec{T} + \kappa v^2 \vec{N}.$$

Tangential and Centripetal Acceleration

Since the tangent unit vector $\vec{T}(t)$ is just the unit vector of $\vec{v}(t)$, representing only the direction of the velocity, we can also write $\vec{v}(t) = v(t)\vec{T}(t)$ (*velocity = speed in the direction of $\vec{T}$*)

From this, we can write a new expression for $\vec{a}(t)$. Since $\vec{a}(t) = \vec{v}'(t)$, then:

$$\vec{a} = \frac{d}{dt}(v\vec{T}) = v'\vec{T} + v\vec{T}'.$$

We have already established in 2.3 that $\vec{T}'$ will always be orthogonal to $\vec{T}$, which is why its unit vector is known as the **normal vector** $\vec{N} = \frac{\vec{T}'}{|\vec{T}'|}$. So, we see that we have broken down $\vec{a}$ into a component pointing tangent to the curve and a component pointing normal to the curve.

Since we are considering vector *components*, a_t and a_c are scalar values. We can now represent $\vec{a}(t)$ using a_t and a_c pointing in their respective directions:

$$\vec{a} = a_t\vec{T} + a_c\vec{N}.$$

We can also rewrite the first equation by using the fact that:

- $\kappa = \frac{|\vec{T}'|}{|\vec{r}'|} \Rightarrow |\vec{T}'| = \kappa|\vec{r}'| = \kappa v$, where κ is the curvature at t
- $\vec{N} = \frac{\vec{T}'}{|\vec{T}'|} \Rightarrow \vec{T}' = |\vec{T}'|\vec{N} = \kappa v\vec{N}$

$$\vec{a} = v'\vec{T} + \kappa v^2\vec{N}$$

Thus, we find that the components of acceleration are:

$$a_t = v' \qquad a_c = \kappa v^2 = \frac{|\vec{r}' \times \vec{r}''|}{|\vec{r}'|^3}.$$

Since curvature is the reciprocal of the radius of curvature R , we can also write:

$$a_c = \frac{v^2}{R}.$$

This is the equation typically introduced in a mechanics course.

2.5 Parametric Surfaces

- A surface can be parameterized, but it must be done so using two variables, as points on the surface can vary independently in two “dimensions”. Similarly, we would be able to parameterize a volume using three variables, and so on.
- Two *parameters*, say, u and v , can independently vary throughout their domains, with each combination producing a point $P(x, y, z)$ on a surface S . Each coordinate of P is a function of both parameters: $P(x(u, v), y(u, v), z(u, v))$. Then the parametric equations of S are:

$$x = x(u, v) \qquad y = y(u, v) \qquad z = z(u, v).$$

- We can also represent a parametric surface using a vector function of two variables. For S as defined above, we can also use $\vec{r}(u, v) = x(u, v)\hat{i} + y(u, v)\hat{j} + z(u, v)\hat{k}$. As u and v vary, the function $\vec{r}$ outputs the position vector for some point on S .

Grid Curves

A method of graphing a parametric surface is by finding the **grid curves** of the surface. Each curve is achieved by setting one of the parameters to a constant value. This results in a parametric equation of only one parameter, which are space curves.

For example, from the surface $\vec{r}(u, v) = \langle u^2 + v^2, u, v \rangle$, we can deduce the following grid curves:

- When $u = 0$, $\vec{r}(v) = \langle v^2, 0, v \rangle$ is the parabola $x = z^2$, opening along the $+x$ -axis
- When $u = k$, $\vec{r}(v) = \langle k^2 + v^2, k, v \rangle$ is the parabola above shifted k^2 units to the right and k units along the $+y$ -axis
- When $v = 0$, $\vec{r}(u) = \langle u^2, u, 0 \rangle$ is a parabola opening to the right on the plane $z = 0$
- When $v = k$, $\vec{r}(u) = \langle u^2 + k^2, u, k \rangle$ is a parabola opening to the right, shifted k^2 units right and k units up, along the $+z$ -axis

Parametric Equations of a Plane

To find the parametric equation of a plane, find a point $\vec{r}_0$ which lies on the plane as well as two nonparallel vectors $\vec{u}$ and $\vec{v}$ which lie on the plane. We can step an arbitrary amount μ and ν in each of the direction of the vectors to reach every point on the plane.

Since the vectors $\vec{u}$ and $\vec{v}$ are linearly independent, they can be considered basis vectors for the plane.

Vector and Parametric Equations of a Plane

Let the plane P contain point $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ and the vectors $\vec{u} = \langle u_1, u_2, u_3 \rangle$ and $\vec{v} = \langle v_1, v_2, v_3 \rangle$ lie on P , and $\vec{u}$ is not parallel to $\vec{v}$.

Then a vector equation of P is

$$\vec{r}(\mu, \nu) = \vec{r}_0 + \mu\vec{u} + \nu\vec{v}.$$

This can be decomposed into the following parametric equations:

$$x(\mu, \nu) = x_0 + u_1\mu + v_1\nu$$

$$y(\mu, \nu) = y_0 + u_2\mu + v_2\nu$$

$$z(\mu, \nu) = z_0 + u_3\mu + v_3\nu.$$